

MATH1860J Recitation Class 02

Foundations, Functions, Convergence: Chapters 06-08

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Outline

1. Recap for Last Week
2. Complex Numbers
3. Functions and Maps
4. Sequences
5. Exercise Solutions

1. Recap for Last Week

- Logic
- Set
- Numbers
- Countability for $\mathbb{Q}$ and $\mathbb{R}$
- Subset of Real Numbers

Limit Inferior and Limit Superior

Definition

Let (a_n) be a bounded real sequence. Define

$$b_n := \inf\{a_k : k \geq n\} \quad \text{and} \quad c_n := \sup\{a_k : k \geq n\}.$$

The **limit inferior** of (a_n) is defined as

$$\liminf_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} b_n = \sup_{n \in \mathbb{N}} \inf\{a_k : k \geq n\}.$$

The **limit superior** of (a_n) is defined as

$$\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} c_n = \inf_{n \in \mathbb{N}} \sup\{a_k : k \geq n\}.$$

Equivalently: $\overline{\lim} U$ is the maximum accumulation point/maximum limit of convergent subsequences in U , $\underline{\lim} U$ is the minimum such limit.

Limit Inferior and Limit Superior

Properties:

- (b_n) is monotonically increasing and (c_n) is monotonically decreasing.
- $\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$.
- (a_n) converges if and only if $\liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n$.

Exercise (1)

Consider the set $U \subset \mathbb{R}$, where $U = A \cup B \cup C$ with

$$A = \{x \in \mathbb{R} : 0 < x \leq 1\},$$

$$B = \{x \in \mathbb{R} : x = 2 - 1/n, n \in \mathbb{N} \setminus \{0\}\},$$

$$C = \{x \in \mathbb{R} : x = -1/n, n \in \mathbb{N} \setminus \{0\}\}.$$

Find $\overline{\lim} U, \underline{\lim} U$. State if one or more of them don't exist.

2. Complex Numbers

Complex Numbers

Problem: The equation $x^2 + 1 = 0$ has no solution in $\mathbb{R}$. $\mathbb{R}$ is algebraically incomplete.



Solution: Introduce a new number i such that $i^2 = -1$. Then the solutions are $x = i$ and $x = -i$. We define addition in $\mathbb{C}$ by

$$(a_1, b_1) + (a_2, b_2) := (a_1 + a_2, b_1 + b_2),$$

so in particular the sum of two real numbers is again a real number. Multiplication is defined as

$$(a_1, b_1) \cdot (a_2, b_2) := (a_1 a_2 - b_1 b_2, a_1 b_2 + a_2 b_1).$$

Neutral element: $(1, 0)$

Definition

The **modulus** or **absolute value** of z is defined by

$$|z| := \sqrt{a^2 + b^2} = \sqrt{z \cdot \bar{z}},$$

Definition

The **open ball** of radius R centered at z_0 is

$$B_R(z_0) := \{z \in \mathbb{C} : |z - z_0| < R\}.$$

A set $S \subset \mathbb{C}$ is called **bounded** if there exists some $R > 0$ such that

$$S \subset B_R(0).$$

3. Functions and Maps

Definition

Let X and Y be sets, and let P be a predicate with domain $X \times Y$. Define

$$f := \{(x, y) \in X \times Y : P(x, y)\}.$$

Assume f satisfies the property:

$$(x_1, y_1), (x_2, y_2) \in f, \quad x_1 = x_2 \implies y_1 = y_2.$$

Define the **domain** of f by

$$\text{dom } f := \{x \in X : \exists y \in Y, (x, y) \in f\}.$$

Then f is called a **function** from $\text{dom } f$ to Y . The set Y is called the **codomain** of f , and the **range** of f is defined by

$$\text{ran } f := \{y \in Y : \exists x \in X, (x, y) \in f\}.$$

Definition

Let $f : X \rightarrow \mathbb{R}$ (or $f : X \rightarrow \mathbb{C}$) be a function. We say that f is **bounded** if there exists a constant $M > 0$ such that

$$|f(x)| \leq M \quad \text{for all } x \in X.$$

- A function is **bounded above** if there exists $M \in \mathbb{R}$ such that $f(x) \leq M$ for all $x \in X$.
- A function is **bounded below** if there exists $m \in \mathbb{R}$ such that $f(x) \geq m$ for all $x \in X$.
- A function is **bounded** if and only if it is both bounded above and bounded below.

Injective, Surjective, and Bijective Functions

Definition

We say that a function $f : X \rightarrow Y$ is

- **injective** if

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

for all $x_1, x_2 \in X$.

- **surjective** if $\text{ran } f = Y$.
- **bijective** if f is both injective and surjective.

Exercise (2)

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions, with $A \neq \emptyset$. Show that:

- a If $g \circ f$ is **injective**, then f is **injective**.
- b If $g \circ f$ is **surjective**, then g is **surjective**.

Definition

Let X be any set. The set X is said to be

- **finite** if there exists a bijective function $f : X \rightarrow \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$.
- **infinite** if it is not finite.
- **countably infinite** if there exists a bijective function $f : X \rightarrow \mathbb{N}$.
- **countable** if there exists an injective function $f : X \rightarrow \mathbb{N}$, and **uncountable** otherwise.

4. Sequences

Definition

Let $\mathcal{I} \subset \mathbb{N}$ be an infinite set. A map $\mathcal{I} \rightarrow \mathbb{R}$ is called a **real sequence**, a map $\mathcal{I} \rightarrow \mathbb{C}$ is called a **complex sequence**, or simply a **sequence**.

- We denote the values of a sequence by a_n , i.e., a sequence maps

$$n \mapsto a_n, \quad n \in \mathcal{I} \subset \mathbb{N}.$$

- For simplicity, we often consider the case $\mathcal{I} = \mathbb{N}$. Generalizations are straightforward.
- Formally, a sequence is the set

$$\{(n, a_n) : n \in \mathcal{I}\}.$$

For simplicity, we write

$$(a_n)_{n \in \mathbb{N}} = (a_n) = (a_0, a_1, a_2, \dots) = a_0; a_1; a_2; \dots$$

The values a_n are also called the **terms** of the sequence.

Definition

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence. A **subsequence** of (a_n) is a sequence of the form $(a_{n_k})_{k \in \mathbb{N}}$, where $(n_k)_{k \in \mathbb{N}}$ is a strictly increasing sequence of natural numbers, i.e.,

$$n_0 < n_1 < n_2 < \cdots$$

- Informally, a subsequence is obtained by selecting infinitely many terms from the original sequence while preserving their order.
- Example: If $a_n = \frac{1}{n}$, then $a_{2k} = \frac{1}{2k}$ is a subsequence.

Definition

A complex sequence $(a_n) : \mathcal{I} \rightarrow \mathbb{C}$, $\mathcal{I} \subset \mathbb{N}$, is said to **converge** with limit $a \in \mathbb{C}$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } n > N \implies a_n \in B_\varepsilon(a),$$

where

$$B_\varepsilon(a) := \{z \in \mathbb{C} : |z - a| < \varepsilon\}$$

is called the ε -neighborhood of a .

Equivalently, we write

$$\lim_{n \rightarrow \infty} a_n = a \quad \text{or} \quad a_n \rightarrow a \text{ as } n \rightarrow \infty.$$

A sequence that does not converge (to any limit) is said to **diverge**.

Definition

A real sequence (a_n) is said to **diverge to infinity** if

$$\forall C > 0, \exists N \in \mathbb{N} \text{ such that } n > N \implies a_n > C.$$

We then write

$$\lim_{n \rightarrow \infty} a_n = \infty \quad \text{or} \quad a_n \rightarrow \infty \text{ as } n \rightarrow \infty.$$

A real sequence (a_n) is said to **diverge to minus infinity** if

$$\forall C < 0, \exists N \in \mathbb{N} \text{ such that } n > N \implies a_n < C.$$

In this case, we write

$$\lim_{n \rightarrow \infty} a_n = -\infty \quad \text{or} \quad a_n \rightarrow -\infty \text{ as } n \rightarrow \infty.$$

Convergence

Lemma

A convergent sequence is bounded.

Lemma

For a sequence (a_n) and $a \in \mathbb{C}$:

$$a_n \rightarrow a \iff a_n - a \rightarrow 0, \quad a_n \rightarrow 0 \iff |a_n| \rightarrow 0.$$

Lemma

A convergent sequence has exactly one limit.

Definition

Let (a_n) and (b_n) be convergent sequences with

$$a_n \rightarrow a \quad \text{and} \quad b_n \rightarrow b, \quad a, b \in \mathbb{C}.$$

Then the following hold:

- ① $\lim_{n \rightarrow \infty} (a_n + b_n) = a + b$
- ② $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = ab$
- ③ $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}$ if $b \neq 0$

Exercise (3)

Prove the third lemma.

Theorem (Squeeze Theorem)

Let (a_n) , (b_n) , and (c_n) be real sequences such that

$$a_n \leq c_n \leq b_n$$

for all sufficiently large n . Suppose that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = a.$$

Then the sequence (c_n) converges, and

$$\lim_{n \rightarrow \infty} c_n = a.$$

Exercise (4)

- 1 Prove that $\lim_{n \rightarrow \infty} n^{1/n} = 1$.
- 2 Prove that $\lim_{n \rightarrow \infty} nq^n = 0$, $|q| < 1$.
- 3 Let (a_n) be a sequence converging to a . Prove that the sequence $b_n = \left(\frac{1}{n} \sum_{j=1}^n a_j \right)$ is convergent with limit a .
- 4 Consider the sequence defined by $a_1 = 2$, $a_{n+1} = 1 + \frac{1}{a_n}$. Show that (a_n) is convergent and find its limit.

Convergence

- ① (2.2.24) Every monotonic and bounded sequence (a_n) converges to its supremum or infimum.
- ② (2.2.30) Let (a_n) be a convergent sequence with $\lim a_n = a$. Then any subsequence of (a_n) is convergent with the same limit.
- ③ (2.2.31) Every real sequence has a monotonic subsequence.
- ④ (2.2.33) If a sequence converges, then the limit is the only accumulation point.
- ⑤ (2.2.35) A number a is an accumulation point of a sequence (a_n) iff there exists a subsequence of (a_n) that converges to a .
- ⑥ (2.2.36) **Bolzano-Weierstrass Theorem:** Every bounded real sequence has an accumulation point.

Reminders

- 1 Get used to $B_\epsilon(a)$.
- 2 When you're dealing with limits, the finite elements don't matter. Things beyond N matter.
- 3 When dealing with some complex sequences, you could consider the squeeze theorem to transform the problem into investigating some easier sequences.
- 4 The terms of sequences are discrete points.
For the properties of continuous functions, you can't take it for granted when dealing with sequences.

5. Exercise Solutions

Solution to Exercise 1

Let

$$U = A \cup B \cup C,$$

where

$$A = (0, 1], \quad B = \left\{ 2 - \frac{1}{n} \mid n \in \mathbb{N}^+ \right\}, \quad C = \left\{ -\frac{1}{n} \mid n \in \mathbb{N}^+ \right\}.$$

- ① $\lim U$ (limit inferior): The accumulation points of U are $[0, 1] \cup \{2\}$. The smallest accumulation point is 0. Hence $\lim U = 0$.
- ② $\overline{\lim} U$ (limit superior): The largest accumulation point is 2. Hence $\overline{\lim} U = 2$.

Equivalently: $\overline{\lim} U$ is the maximum limit of convergent subsequences in U , $\lim U$ is the minimum such limit.

Solution to Exercise 2

Proof.

(a) Suppose $g \circ f$ is injective. To show f is injective, assume $f(x_1) = f(x_2)$ for some $x_1, x_2 \in A$. Then

$$(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2).$$

Since $g \circ f$ is injective, we have $x_1 = x_2$. Therefore, f is injective.

(b) Suppose $g \circ f$ is surjective. To show g is surjective, let $c \in C$ be arbitrary. Since $g \circ f$ is surjective, there exists $a \in A$ such that $(g \circ f)(a) = c$. Let $b = f(a) \in B$. Then

$$g(b) = g(f(a)) = (g \circ f)(a) = c.$$

Therefore, g is surjective. □

Solution to Exercise 3

Proof.

We need to prove: If $a_n \rightarrow a$ and $b_n \rightarrow b$ with $b \neq 0$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}$.

Step 1: Show that $\frac{1}{b_n} \rightarrow \frac{1}{b}$.

Since $b \neq 0$, there exists N_1 such that for all $n > N_1$, we have $|b_n - b| < \frac{|b|}{2}$. Thus

$$|b_n| \geq |b| - |b_n - b| > |b| - \frac{|b|}{2} = \frac{|b|}{2}.$$

For any $\varepsilon > 0$, there exists N_2 such that for all $n > N_2$,
 $|b_n - b| < \frac{|b|^2 \varepsilon}{2}$.



Solution to Exercise 3 (continued)

Proof (continued).

For $n > \max\{N_1, N_2\}$:

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b - b_n|}{|b_n| \cdot |b|} < \frac{|b|^2 \varepsilon / 2}{(|b|/2) \cdot |b|} = \varepsilon.$$

Step 2: Apply the product rule.

Since $a_n \rightarrow a$ and $\frac{1}{b_n} \rightarrow \frac{1}{b}$, by the product rule (Lemma 2 from the convergence properties):

$$\frac{a_n}{b_n} = a_n \cdot \frac{1}{b_n} \rightarrow a \cdot \frac{1}{b} = \frac{a}{b}.$$



Solution to Exercise 4(1)

Proof.

Let $a_n = n^{1/n}$. For $n \geq 2$, we have $a_n > 1$. Write $a_n = 1 + h_n$ where $h_n > 0$.

By the binomial theorem:

$$n = a_n^n = (1 + h_n)^n \geq 1 + nh_n + \frac{n(n-1)}{2}h_n^2.$$

Therefore:

$$\frac{n(n-1)}{2}h_n^2 \leq n-1 \implies h_n^2 \leq \frac{2(n-1)}{n(n-1)} = \frac{2}{n}.$$

Thus $0 < h_n \leq \sqrt{\frac{2}{n}}$. Since $\sqrt{\frac{2}{n}} \rightarrow 0$ as $n \rightarrow \infty$, by the squeeze theorem, $h_n \rightarrow 0$.

Therefore, $n^{1/n} = 1 + h_n \rightarrow 1$. □

Solution to Exercise 4(2)

Proof.

Since $|q| < 1$, we can write $|q| = \frac{1}{1+r}$ for some $r > 0$.

By the binomial theorem, for $n \geq 2$:

$$(1+r)^n = \sum_{k=0}^n \binom{n}{k} r^k \geq \binom{n}{2} r^2 = \frac{n(n-1)}{2} r^2.$$

Therefore:

$$|nq^n| = n|q|^n = \frac{n}{(1+r)^n} \leq \frac{n}{\frac{n(n-1)}{2} r^2} = \frac{2}{(n-1)r^2}.$$

Since $\frac{2}{(n-1)r^2} \rightarrow 0$ as $n \rightarrow \infty$, by the squeeze theorem:

$$\lim_{n \rightarrow \infty} nq^n = 0.$$



Solution to Exercise 4(3)

Proof.

Let $a_n \rightarrow a$. We prove that $\left(\frac{1}{n} \sum_{j=1}^n a_j\right) \rightarrow a$.

Let $\varepsilon > 0$. Since $a_n \rightarrow a$, there exists N such that for all $j > N$, $|a_j - a| < \frac{\varepsilon}{2}$.

For $n > N$, write:

$$\left| \frac{1}{n} \sum_{j=1}^n a_j - a \right| = \left| \frac{1}{n} \sum_{j=1}^n (a_j - a) \right| \leq \frac{1}{n} \sum_{j=1}^n |a_j - a|.$$



Solution to Exercise 4(3) (continued)

Proof (continued).

Split the sum:

$$\frac{1}{n} \sum_{j=1}^n |a_j - a| = \frac{1}{n} \sum_{j=1}^N |a_j - a| + \frac{1}{n} \sum_{j=N+1}^n |a_j - a|.$$

The first term: $\frac{1}{n} \sum_{j=1}^N |a_j - a| \leq \frac{NM}{n} < \frac{\varepsilon}{2}$ for sufficiently large n , where

$$M = \max_{1 \leq j \leq N} |a_j - a|.$$

The second term: $\frac{1}{n} \sum_{j=N+1}^n |a_j - a| < \frac{n-N}{n} \cdot \frac{\varepsilon}{2} < \frac{\varepsilon}{2}.$



Solution to Exercise 4(4)

Proof.

Consider $a_1 = 2$, $a_{n+1} = 1 + \frac{1}{a_n}$.

Step 1: Show that (a_n) is bounded. We claim $1 < a_n < 2$ for all $n \geq 2$ (by induction).

Base: $a_2 = 1 + \frac{1}{2} = \frac{3}{2} \in (1, 2)$.

Step: If $1 < a_n < 2$, then $a_{n+1} = 1 + \frac{1}{a_n} \in (1 + \frac{1}{2}, 1 + 1) = (\frac{3}{2}, 2)$.

Step 2: Suppose (a_n) converges to L . Then taking limits in $a_{n+1} = 1 + \frac{1}{a_n}$:

$$L = 1 + \frac{1}{L} \implies L^2 - L - 1 = 0 \implies L = \frac{1 + \sqrt{5}}{2} \text{ (golden ratio).}$$

Step 3: To show convergence, observe that odd/even subsequences are monotonic and bounded, hence converge (by Theorem 2.2.24). Both must converge to the same limit by the recurrence relation. □

Thanks for Listening!

Any questions? Please feel free to ask!

References:

- Horst Hohberger, *math186_all_lecture_slides (2026)*
- Wenxin Chen, *MATH1860J RC*, 25 Fall
- Leqi Tang, *MATH1860J RC*, 25 Fall
- Zhun Wang, *MATH1860J RC*, 25 Fall

To be continued...

- Metric Spaces
- Cauchy Sequences

Enjoy the National Holiday!